

JPEG AI as a Threat to Deepfake Detection

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Outline

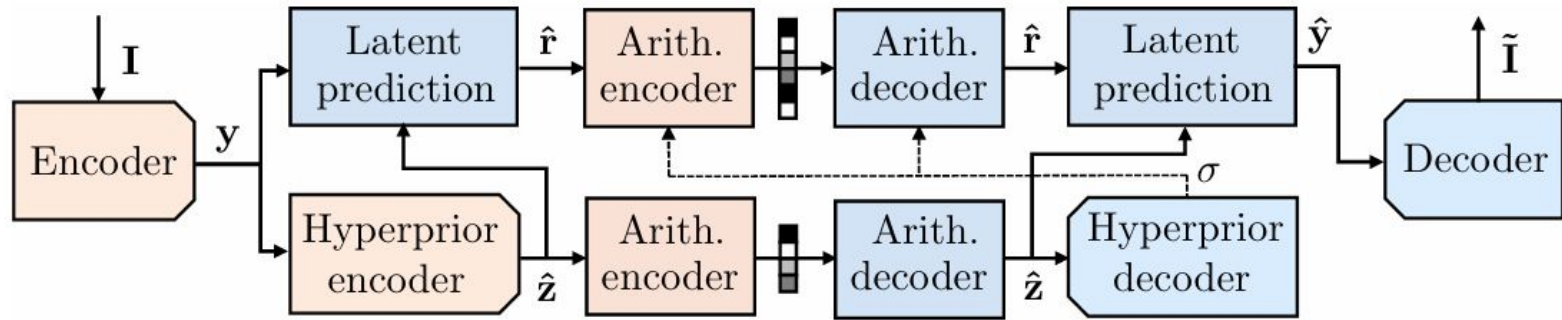
- I. AI JPEG
- II. Side effect of compression
- III. Current available solutions
- IV. Possible solutions
- V. Final consideration

What is JPEG?

Joint Photographic Experts Group (JPEG):

- Compression standard to reduce a image file size
- Uses DCT to removes less meaningful information (High freq) from the image
- The compression is lossy

JPEG AI compression



Usage of NN to compress the image, in order to obtain a result of higher visual quality for the same bit per pixel rate, while benefiting from the typical advantages of a neural network

Deep Fakes

A form of synthetic media, made by editing existing media or created completely, through the usage of AI tools or other software.

Some examples:

- Face Swap
- Age Transformation
- Expression Manipulation
- etc.

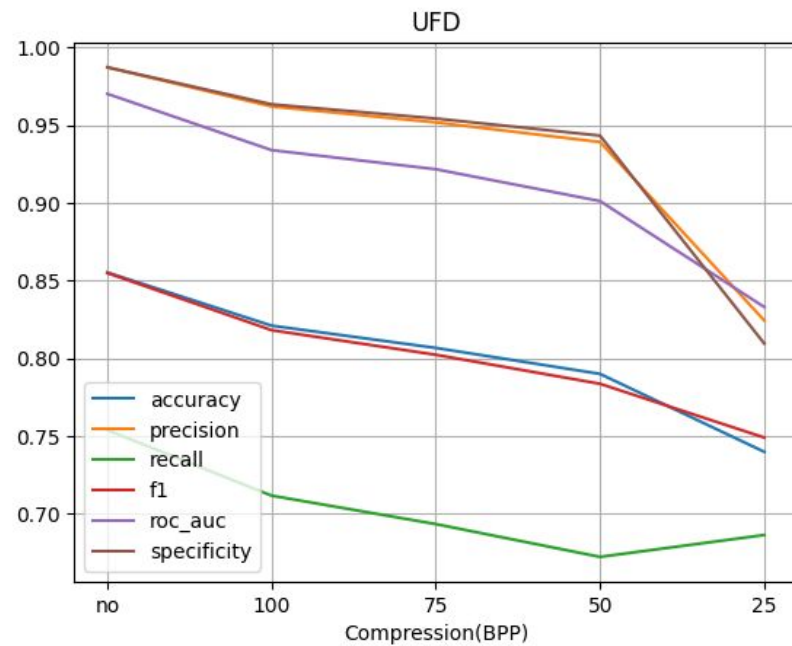
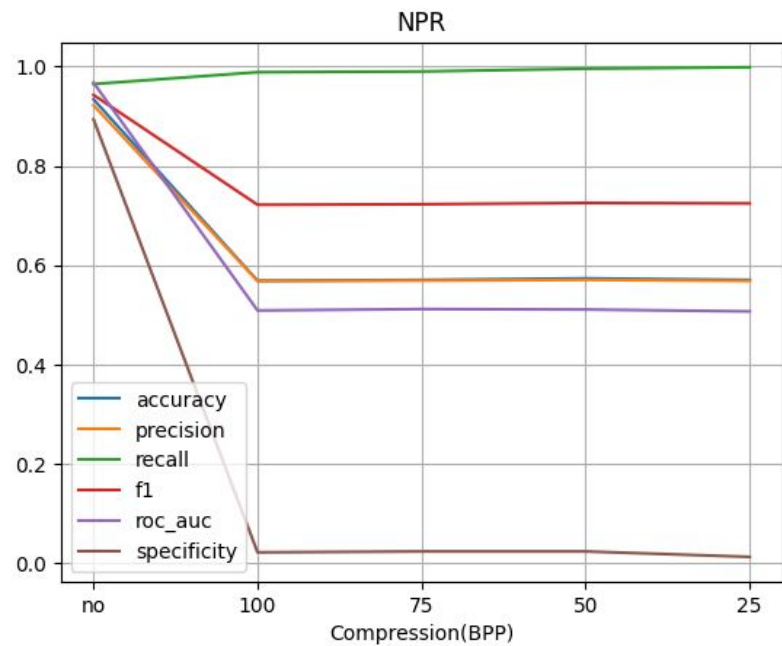


Deep Fakes detection

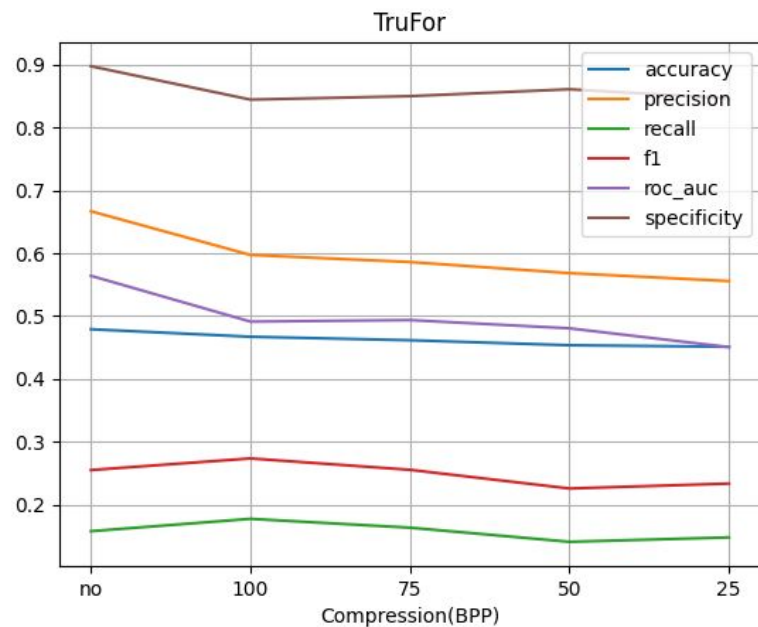
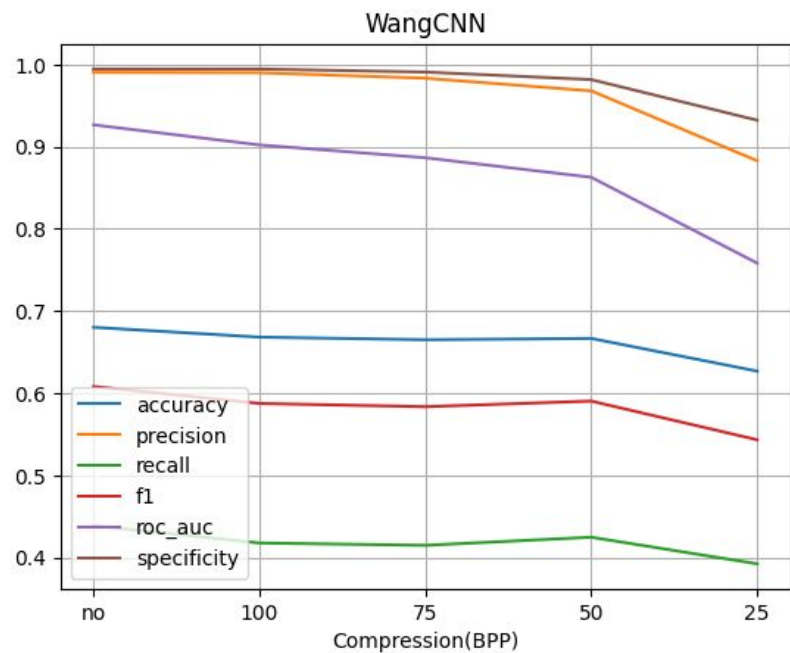
While the initial iteration of image generators produce easily spottable results, nowadays it's not so easy, and will likely get harder.

The solutions is to create AI tools to detect the creations of other AI tools

Effects on fake detection (1/3)



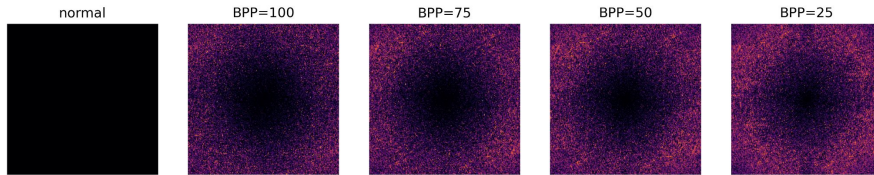
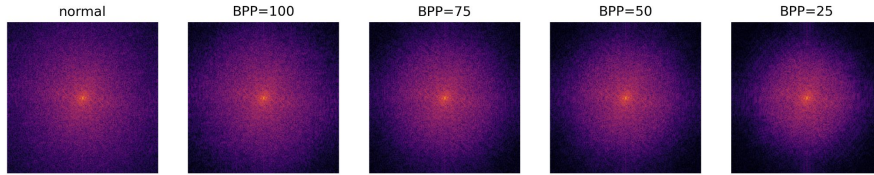
Effects on fake detection (2/3)



Effects on fake detection (3/3)

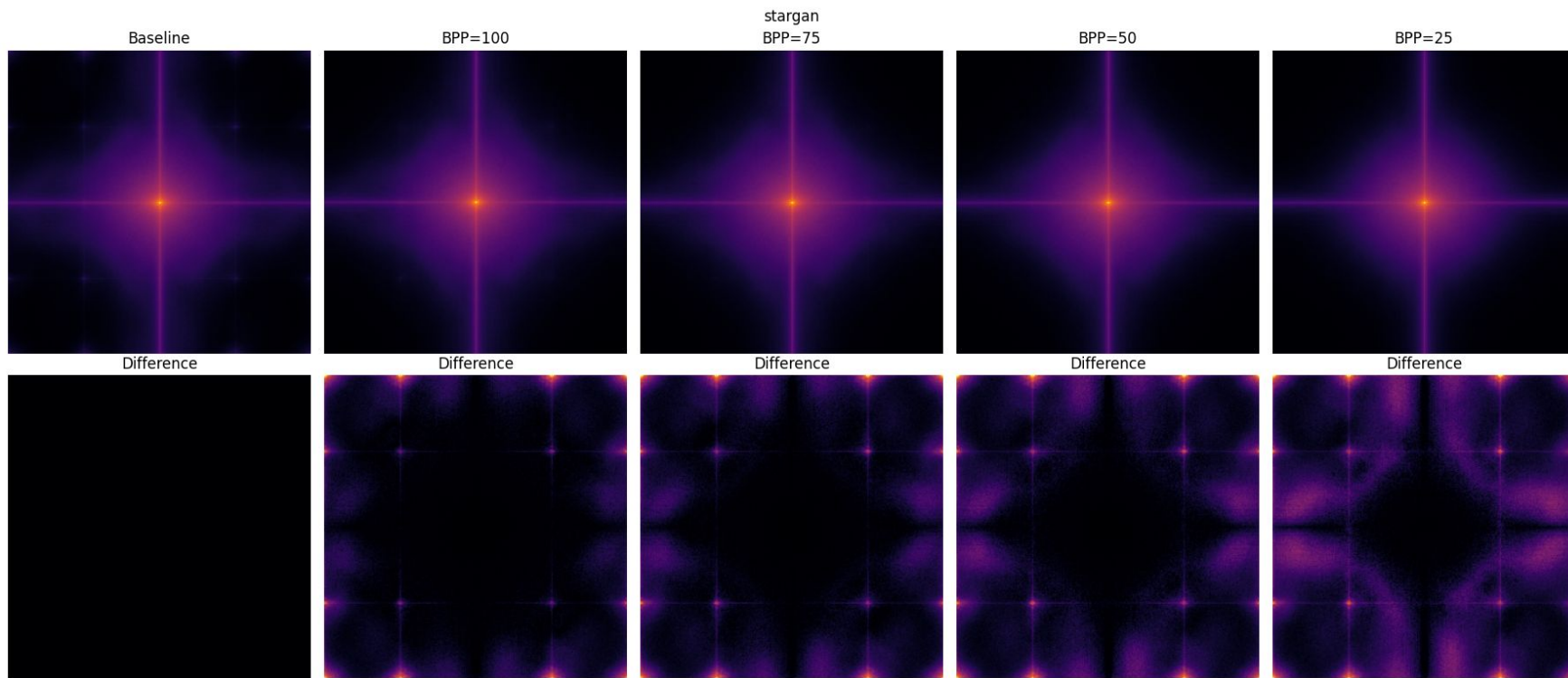
	NPR	UFD	WangCNN	TruFor
BASELINE	0.933	0.855	0.680	0.478
BPP=1.00	0.568	0.821	0.668	0.466
BPP=0.75	0.570	0.806	0.665	0.461
BPP=0.50	0.573	0.789	0.666	0.453
BPP=0.25	0.570	0.739	0.626	0.451

A closer look to the effect on the image (1/4)

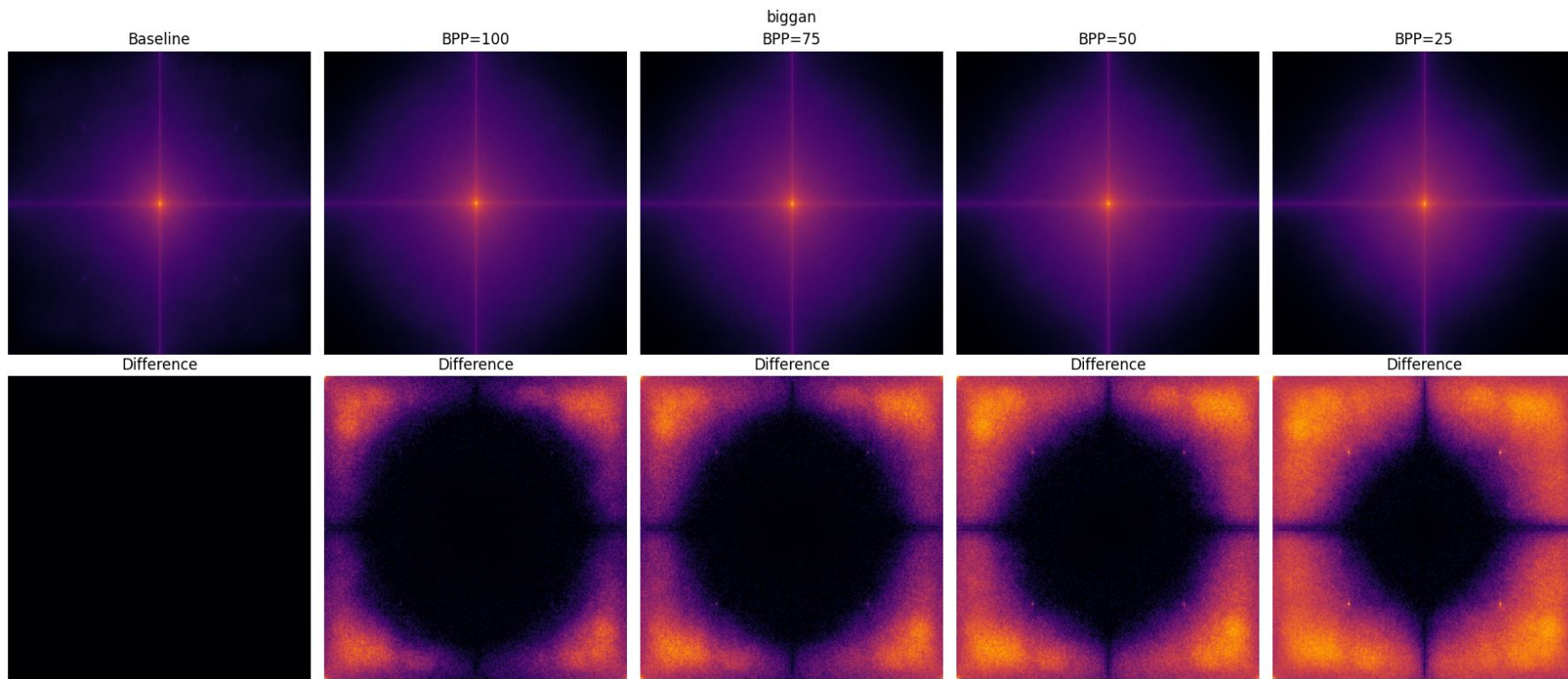


While the overall image looks the same, compression modifies the details, as shown by the fact that the increasing difference in the high frequencies zone

A closer look to the effect on the image (2/4)

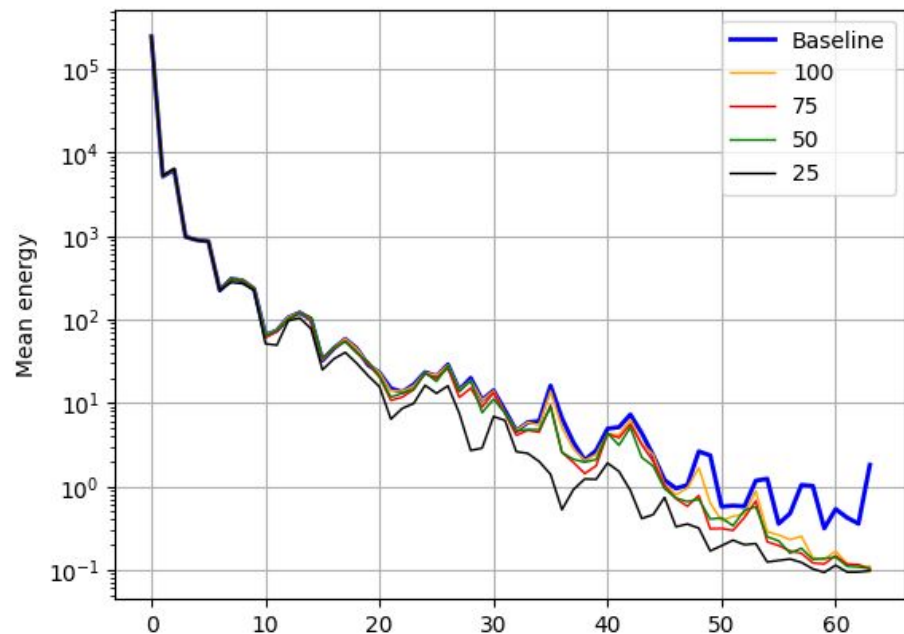


A closer look to the effect on the image (3/4)

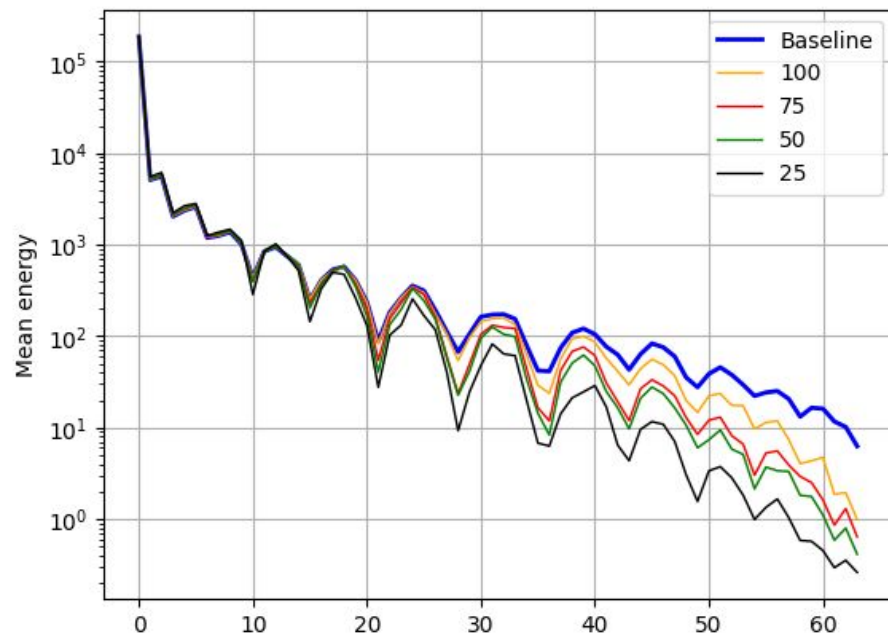


A closer look to the effect on the image (4/4)

Fake image



Real image



Current solutions

Because of the fact that the problem is very recent, no definitive solution has been found yet

So far the only methods that have been proposed are to fine tune the models or retrain them from scratch, in both cases augmenting the dataset with compressed images

Possible solutions

Given the intense effort needed to retrain a model from zero, especially in the case of big networks with large datasets, it would be of interest to improve the performance obtained from fine tuning.

Here are proposed two methods to do so:

- Modifying the classifier head to have four classes, making a distinction between compressed and uncompressed images
- Using the average spectrum loss in compression to weight the image in preprocessing

Experimental setup (Dataset)

For dataset CNNSynth and TinyGenImage have been used as basis, filtered for 256x256 images.

The training set is composed by roughly 60k images, augmented with 4 levels of AI compression mixed together.

Validation and test sets are 10k and 6k divided by compression quality in smaller subsets

Experimental setup (Training)

We take NPR as our base model because it's the one that suffers the worst performance degradation.

The training is divided in two stages, the first with a learning rate of 10^{-3} for the classifier head and a second one with a learning rate of 10^{-5} for the whole network.

The training implements an early stop logic, with a patience of 8 for the first phase and 4 for the second. The maximum number of iterations is 50

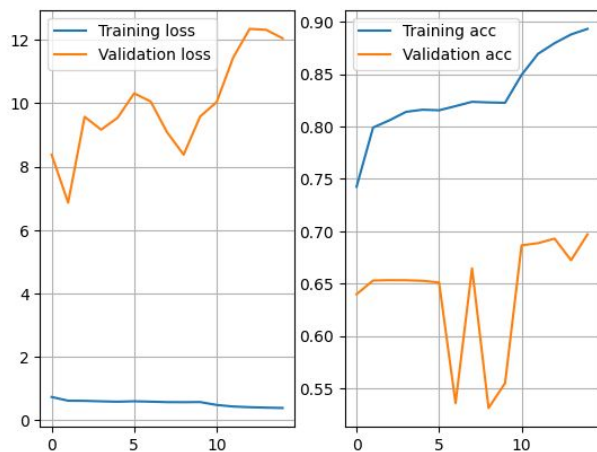
Early stops keeps track of the average accuracy across the validation subsets

Results (1/3)

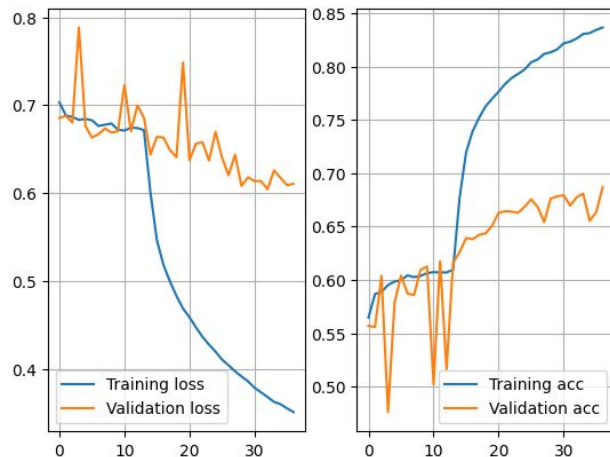
	BASELINE	BPP=1.00	BPP=0.75	BPP=0.50	BPP=0.25
NPR	0.933	0.568	0.570	0.573	0.570
NPR FT	0.981	0.640	0.630	0.642	0.607
NPR 4C	0.934	0.640	0.626	0.620	0.625
NPR PP	0.870	0.642	0.615	0.599	0.610

Results (2/3)

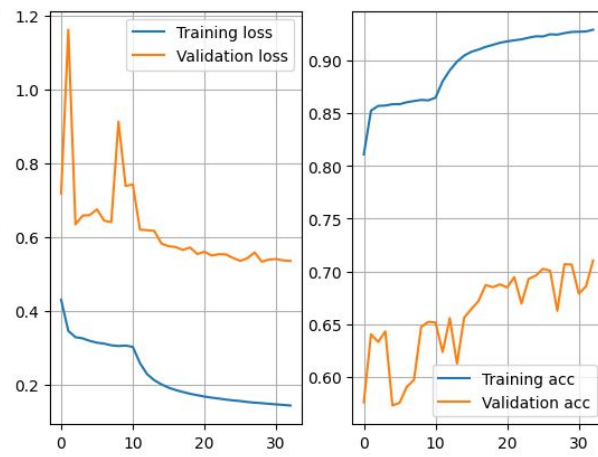
4 Classes



Modified preprocessing

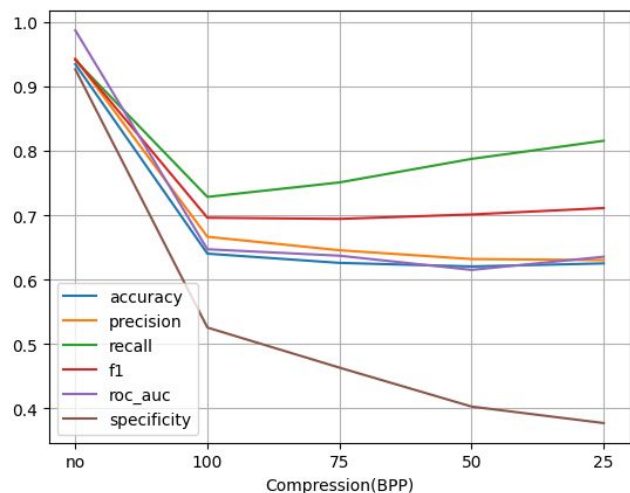


Control Fine Tuning

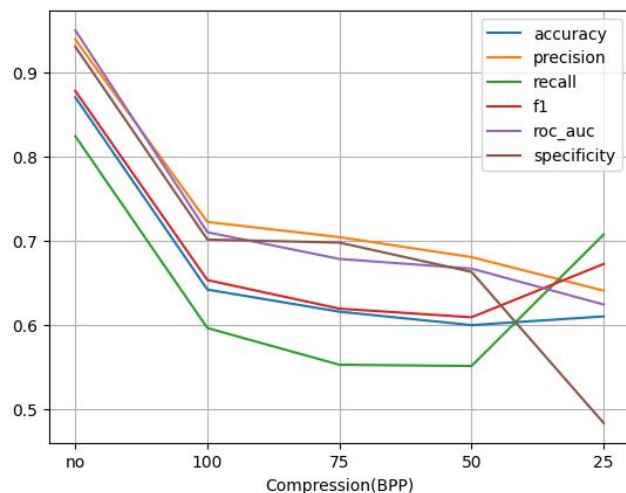


Results (3/3)

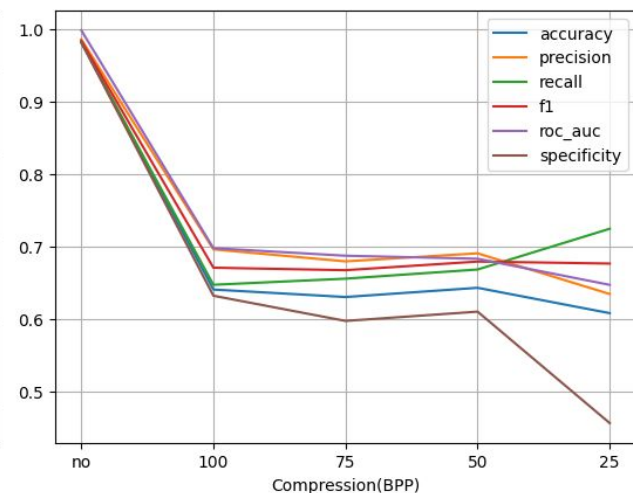
4 Classes



Modified preprocessing



Control Fine tuning



References

Is JPEG AI Going to Change Image Forensics? [E. D. Cannas et al., 03/2025]

Thanks you for the attention